注意：

允許學生個人、非營利性的圖書館或公立學校合理使用本基金會網站所提供之書籍、各項試題及其解答。**可直接下載而不須申請**。

重版、系統地複製或大量重製這些資料的任何部分，必須獲得財團法人臺北市九章數學教育基金會的授權許可。

申請此項授權請電郵 chiuchang.tw@gmail.com

Notice:

Individual students, nonprofit libraries, or schools are permitted to make fair use of the books, papers and its solutions. Republication, systematic copying, or multiple reproduction of any part of this material is permitted only under license from the Chiuchang Mathematics Foundation.

Requests for such permission should be made by e-mailing Mr. Wen-Hsien SUN chiuchang.tw@gmail.com

International Mathematics Assessments for Schools (2024-2025)

Upper Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

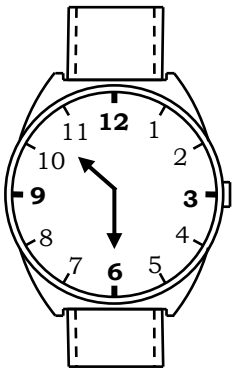
The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Upper Primary Division Round 2

Questions 1 to 5, 4 marks each

1. Compared to a standard clock, an old clock runs 9 seconds faster every hour, while an old watch runs 6 seconds slower every hour. At 10:30:00 AM today, they show the same correct time, as shown in the diagram below.

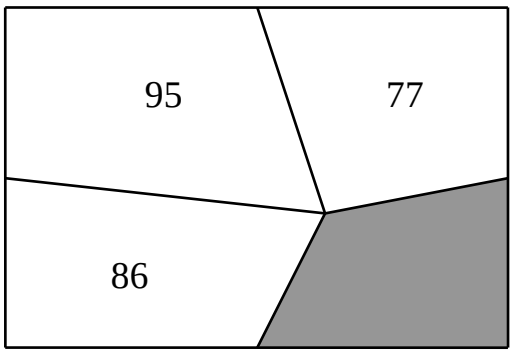
After today, in how many days will this old clock and old watch show the same correct time again at 10:30:00 AM?

（A）120　　（B）200　　（C）300　　（D）450　　（E）600

Answer：_____

---

2. Inside a rectangle, a certain point has been chosen. By connecting this point with the midpoints of the four sides of the rectangle, the rectangle is split into four disjoint regions. It is known that the areas of three of these regions are 95 cm^2, 86 cm^2 and 77 cm^2 respectively, as shown in the diagram below.



What is the area, in cm^2, of the shaded region?

Note: The diagram is NOT drawn to scale.

（A）67　　（B）68　　（C）86　　（D）87　　（E）104

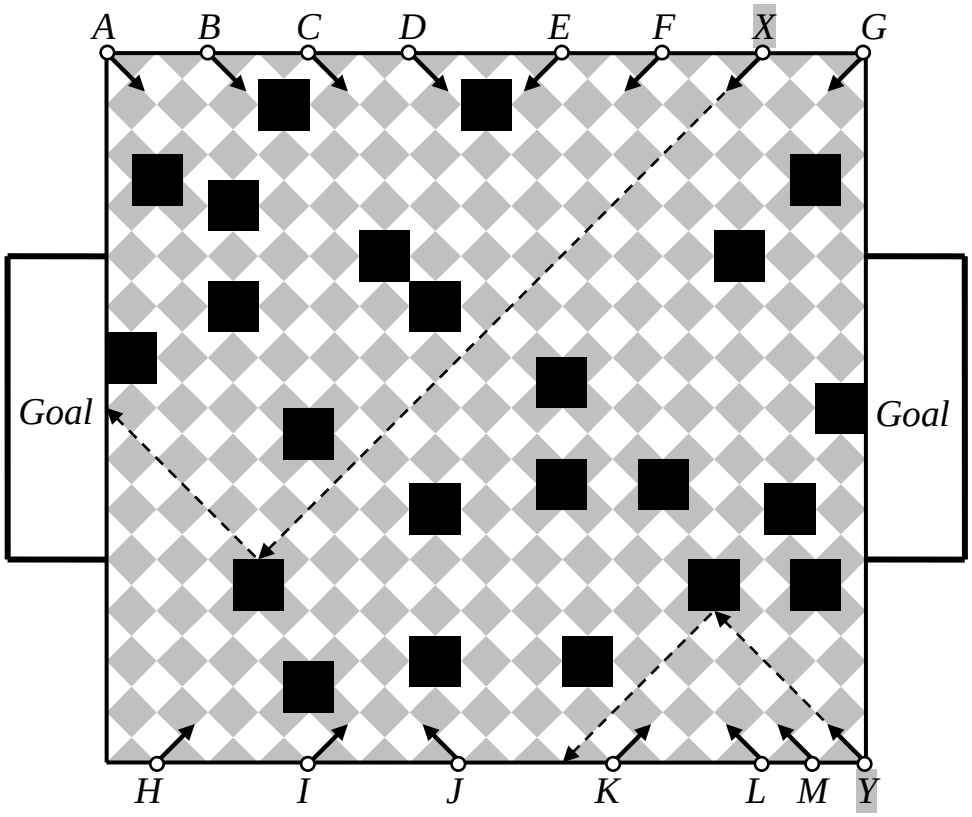
Answer：_____

3. Ann bought a total of 17 blue and red pens for \$222. Each pen costs an integer number of dollars, and a red pen costs \$3 more than a blue pen. How many red pens did Ann buy?

（A）6　　（B）7　　（C）8　　（D）11　　（E）12

Answer：_____

4. A game board is shown in the diagram below. 15 balls are placed at the points indicated by the letters and will be shot in the directions indicated by the arrows. The black squares are obstacles such that when each ball hits the obstacle, it will be reflected by its path, where the angle of incidence is equal to the angle of reflection. For example, when the ball located at point *X* is shot, it would end up in the goal on the left (as shown by the dotted lines); while when the ball located at point *Y* is shot, it will eventually go outside (as shown by the dotted lines). The balls cannot be reflected from any border of the game board.

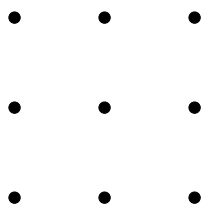


If the balls located at points *A*, *B*, *C*, *D*, *E*, *F*, *G*, *H*, *I*, *J*, *K*, *L* and *M* are shot one after another in the directions indicated by their corresponding arrows, how many balls will eventually end up in one of the goals, **without** counting the example (the ball starting from *X*)?

（A）4　　（B）5　　（C）6　　（D）7　　（E）8

Answer：_____

5. There are nine grid points in a 2×2 table shown below.



How many different lines can we draw on the grid which passes through exactly two grid points?

（A）10　　（B）12　　（C）16　　（D）18　　（E）24

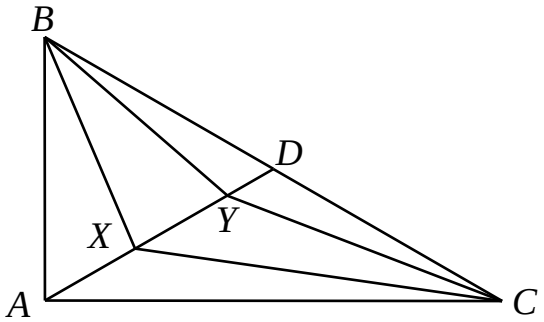
Answer：_____

Questions 6 to 13, 5 marks each

6. A string, which has an integer length, in cm, is to be cut into 4 pieces. It is known that the lengths of all 4 pieces are all different integers. If the length of each piece is $\frac{1}{2}$ times the length of the previous piece plus 1 cm, except for the longest piece, what is the shortest possible length, in cm, of the string?

Answer：_____ cm

7. In triangle ABC, $\angle ACB = 30°$ and $\angle BAC = 90°$, where D is a point on BC. Points X and Y are on AD such that the areas of triangles AXB, AXC, XBY, XCY and YBC are all equal, as shown in the diagram below.



If $AB = 25$ cm, what is the length, in cm, of XY?

Answer：_____ cm

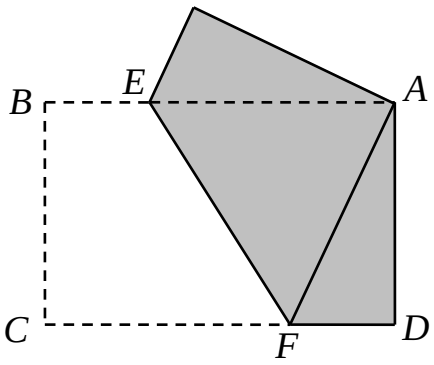
8. Select nine different digits from 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9, and then use these digits to create three positive three-digit numbers, namely A, B and C, where $A > B > C$. What is the smallest possible sum of the pairwise positive differences (subtracting the smaller number from the larger number, that is $A-B$, $A-C$ and $B-C$) between these three three-digit numbers?

Answer：_____

9. Alex and Boris live in the same village. One day, both of them left their village at the same time to travel to the city, where it is known that they have different constant speeds. At a certain moment in the journey, we know that if Alex had gone three times as far as he had so far, he would have half as far to go, and if Boris had gone half as far as he had so far, he would have three times as far to go. How many times is Boris faster than Alex?

Answer : _____

10. The area of a rectangular paper *ABCD* is 120 cm^2. Now, fold this paper along line *EF* so that *C* coincides with *A*, where *E* is on *AB* and *F* is on *CD*, as shown in the diagram below. If $FD = \frac{1}{3}CD$, what is the area, in cm^2, of the shaded region?



Answer : _____ cm^2

11. Amy selects one number from 1, 2, 3, …, 99 and 100. After that, Brandon selects a number from the remaining ninety-nine numbers such that the last digit of the product of their numbers is 4. How many ways are there for Amy and Brandon to select their numbers?

Answer : _____ ways

12. Excluding the cue ball (white ball), a complete set of snooker balls consist of 21 balls, of which there are 15 red balls, which are worth 1 point each, and the 6 other non-red balls which are worth 2, 3, 4, 5, 6 and 7 points respectively. Two balls are randomly selected from these 21 balls at the same time. Let *p* be the probability that the total value of the two balls selected is at most 5 points. What is the value of 210*p*?

Answer : _____

13. If a positive integer, composed of at least three digits, has all its digits written in strictly decreasing order from left to right, then this positive integer is called a ***good*** integer. For example, 876 and 97531 are both ***good*** integers, while each of 87, 5331 and 97513 are not. How many ***good*** integers are there in total?

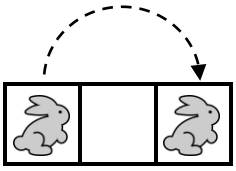
Answer : _____

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

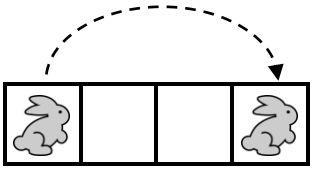
14. Let a, b and c be digits so that the decimal number $\overline{2a41b3c125}$ is a multiple of 7, 9 and 11. List down all the possible numerical values of $\overline{2a41b3c125}$.

Answer : _____

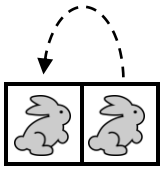
15. In one move, a bunny can either jump forward 2 or 3 steps or jump backward 1 step, as shown in the diagrams below.



Jump forward 2 steps

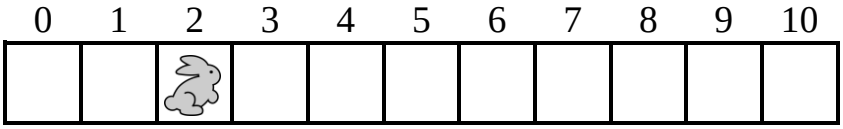


Jump forward 3 steps



Jump backward 1 step

Starting from cell number 2, in how many different ways can the bunny reach cell number 8 in less than or equal to 5 moves? Note that the different order of jumping forward 2 or 3 steps and backward 1 step are viewed as different ways. Also, the bunny is only allowed to jump on any cell from 0 to 10.



Answer : _____ ways